

UNIT-02

1. Software Process [AKTU-2020-21]
2. Process Models
3. Choice of Process Model [AKTU-2020-21]
4. Rapid Application development [AKTU-2020-21, 22-23]
5. Agile method
6. Dynamic Sys^m Development Method
7. Extreme Prog^m Managing interactive Process
8. Basics of Software estimation
9. COSMIC Full Funⁿ Points [AKTU-2022-23]
10. Cocomo II
11. Productivity Model

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Software Process Model :- [IMP]

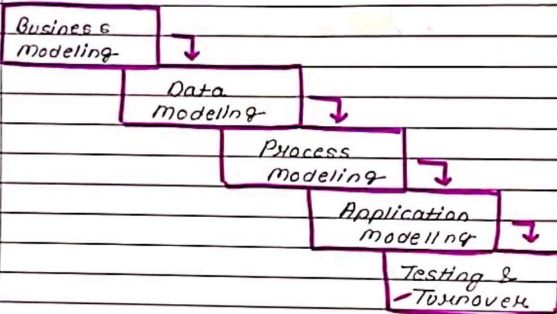
- Representation of Software Process
- A Software Process Model is a structured approach that defines the steps and activities involved in developing and maintaining software system

Choice of Process model :- [IMP]

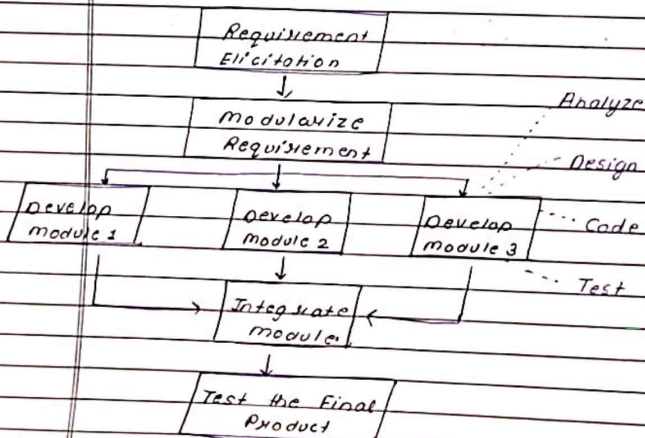
- Project Requirement
- Project Size • Complexity
- Client
- Skills and knowledge
- Developer & team member

Rapid Application Development :- [IMP]

- Quick Prototyping
- Iterative Development
- Incremental Releases
- User Involvement
- Time - Boxing
- Parallel Development
- Project can be broken down into small module where each module can be assigned independently to separate teams



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Advantage :-

- Reduce development time
- Reusability

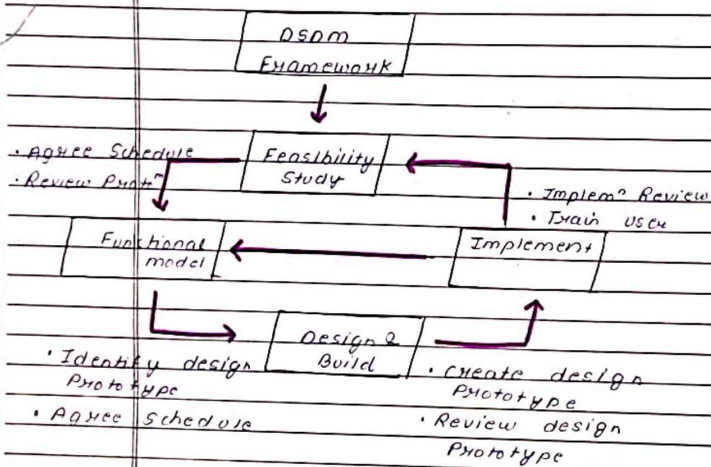
Disadvantage :-

- Not Suitable For Smaller Project
- High skill Person Required

DSDM (Dynamic System Design Development Model)

- Work on same principle of Agile development, including Continuous User / Customer involvement
- DSDM is a Framework largely based around Rapid Application development. It focuses on Information System Projⁿ that are characterized by tight Schedules and budgets
- The method Primary aim is to deliver business needs and real time benefits also DSDM also make sure that benefits are clear, has Feasible Solⁿ with Solid Foundation already in place before a Project is Started

OSDM Life Cycle



Four Phase :-

- 1 Feasibility and business study
- 2 Functional model Prototype iteration
- 3 Design and build iteration
- 4 Implementation

1 Feasibility Study

Check Project is suitable for OSDM by understanding the requirements & constraints

2 Business Study

- Define user & Functional requirements necessary for application to provide business value
- Also identify how the application is designed & maintained in long time/sun

3 Functional Model Iteration

- Create Prototype - get feedback from user to gather more requirement & make improvements

4 Design and Build Iteration

Refine the Prototype

5 Implementation

Put the latest version of the SW into real life working environment

Extreme Programming :-

Good Practices need to be Practical in Extreme Programming

• Code Review :-

- Detect and correct errors efficiently
- Carried out by Pair of Programmer

• Testing

- Help to remove errors and increase reliability

- Suggest TDD (Test-driven develop^m)
Test Cases are written even before any code is written.

• ~~Target~~

Basics of Software Estimation:

- Effective Software Project Estimation is important activity in Software development Project
- It Predicts the time and budget that required for completing a Project
- Project Estimation requires the use of complex tools & good mathematical as well as knowledge about Planning
- The main reasons Software Pg Fail is our inability to accurately estimate Software size

Responsible Persons

- Software Manager
- Cognizant Engineers
- Software Estimators

User Requirement

Constraints

Organizational Policies

Software Project Estimation

Schedule

Efforts

Cost

Factor affect on Project Estimation:-

1. Cost
2. Time
3. Size & Scope
4. Risk
5. Resources

Steps of Software Project Estimation:

1. Estimate Project Size
2. Estimate Efforts
3. Estimate Project Schedule
4. Estimate Project Cost

COSMIC Full Function Points :-

Effort and Cost Estimation Technique :-

- Top down Estimation &
- Bottom up Estimation

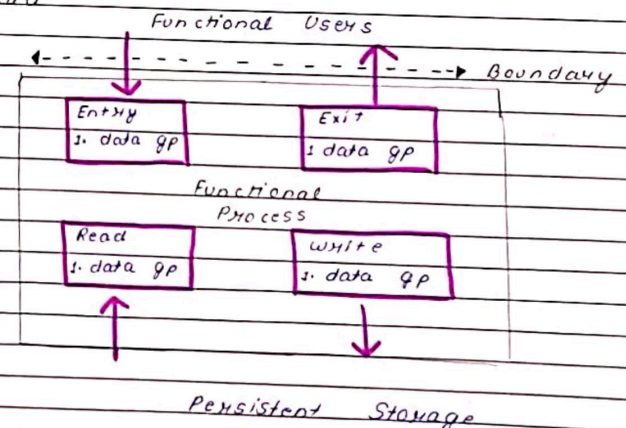
- Expert Judgement
- Analogous Estimation

Cosmic Function Points [Imp]

- Cosmic Funⁿ Point is a Software matrix Used to measure the Size & Complexity of a Software System
- It quantifies the Functionality of the System rather than Focusing on line of code or technical details
- Features and Functionalities of the Software are assigned weight based on their Complexity & impact
- Total Cosmic Funⁿ Points are calculated by Summing up the weighted Values
- CFP helps estimate the efforts and resources required for Sof^m dev^m.
- It Provides a Standardized measure for Comparing and estimating the Size & Complexity of different Software System

Uses of Cosmic Funⁿ Points

- Estimate development effort
- Estimate Project duration
- Estimate Project quality
- Estimate test effort
- An approximation for measurement purposes, the Cosmic method assumes that the Functionality of any data manipulation is accounted for by the data movement
- There are four data movement types
 1. Entry
 2. Exit
 3. Write
 4. Read



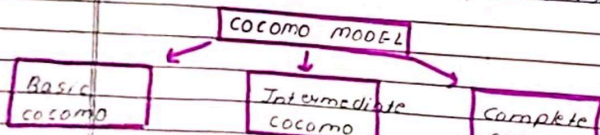
COCOMO MODEL I

- Estimate Effort Cost based on size
- Less Accurate
- Limited Customization

COCOMO MODEL II

- Estimate Cost based on Project Complexity, team capabilities.
- More Accurate
- Feasibility & Customization

- COCOMO Stands For "Constructive Cost Model". It is one of the Very Famous model which is used to estimate the Cost of the Project
- COCOMO model was Proposed by Boehm in the Year 1981
- Using this model we can estimate the time (months) and no. of people needed to develop the Projⁿ



Parametric Productivity Model :-

- Mathematical model estimate how different factor affects the efficiency of Software development. It consider Variable like - team experience, Project Complexity & tools to Predict the Productivity ~ levels

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